

AWS S3 STORAGE

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Project: AWS S3 Storage Setup

Objective

- Login AWS
- Project purpose
- Create S3 Bucket
- Store and Manage Files

Architecture Diagram

- Picture
- Explanation

Resource Created

- S3 Bucket Creation
- Folder Creation
- File Upload
- Object Management

Configuration Details

- Bucket Name
- AWS Region
- Block Public Access
- Versioning
- Default Encryption
- Tags (optional)

Security Rules

- IAM User/Policy
- Bucket Policy
- ACL (Access Control List)
- Object Ownership

Storage Management

- Upload Objects
- Download Objects
- Delete Objects
- Folder Management

- Storage Classes

Data Protection

- Versioning
- Lifecycle Rules
- Replication (CRR/SRR)
- Backup & Recovery

Testing Results

- Upload File Test
- Download File Test
- Public Access Test
- Bucket Policy Test
- Bucket Created Successfully
- Files Uploaded Successfully
- Access Verified

Conclusion

- Project Summary
- Learning Outcome

1. OBJECTIVE -

- Purpose: Login to AWS Management Console using IAM User or Root User Credentials Store files Securely.

Steps:

1. Login to AWS Management Console.
2. Search Amazon S3 Bucket in the search bar.
3. Open the Amazon S3 Bucket Dashboard.

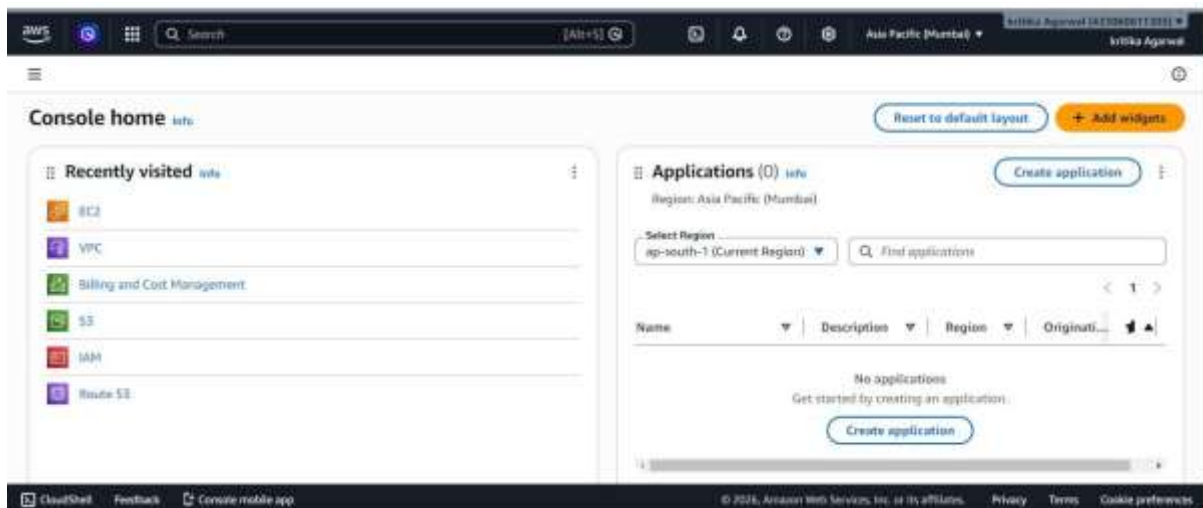


FIG-1: AWS CONSOLE HOME PAGE

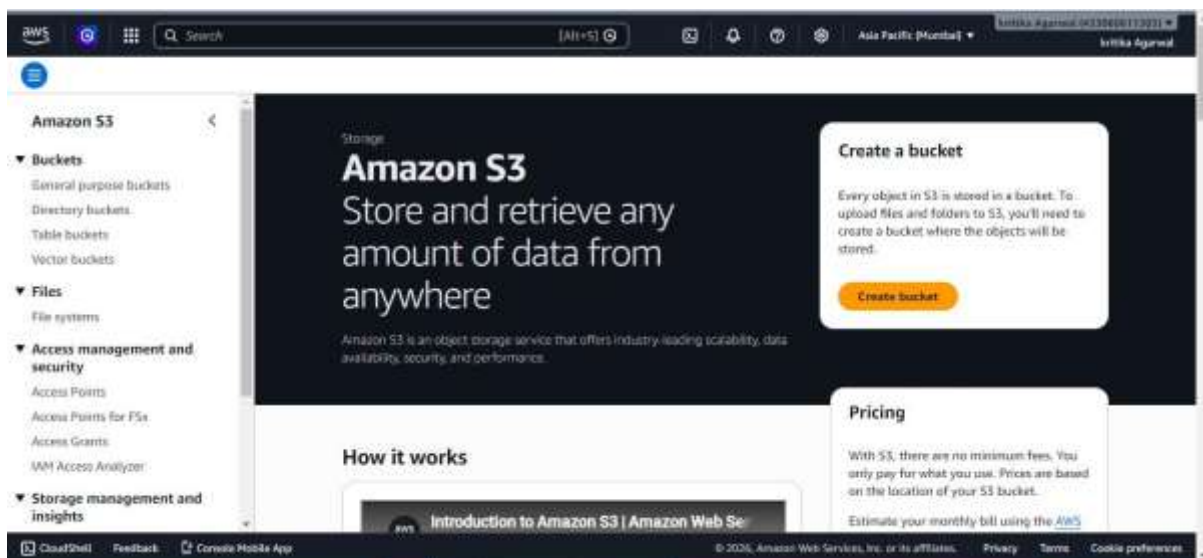


FIG-2: AMAZON S3 BUCKET DASHBOARD

2. ARCHITECTURE DIAGRAM-

The architecture diagram represents the overall workflow of Amazon S3 (Simple Storage Service). It shows how a user accesses the AWS Management Console, creates an S3 bucket, uploads files (objects), and securely stores data in the cloud. The architecture also includes security features such as IAM permissions, Bucket Policies, Block Public Access, Encryption, and Versioning. This diagram helps in understanding how Amazon S3 stores, manages, protects, and provides access to data in a secure and scalable manner.

Description: This architecture contains-

- User
- AWS Management Console

- Amazon S3 Bucket
- Objects (Files)
- Folders (Prefixes)
- IAM User/Role
- Bucket Policy
- Block Public Access
- Versioning
- Server-Side Encryption
- Storage Classes
- AWS Cloud Infrastructure

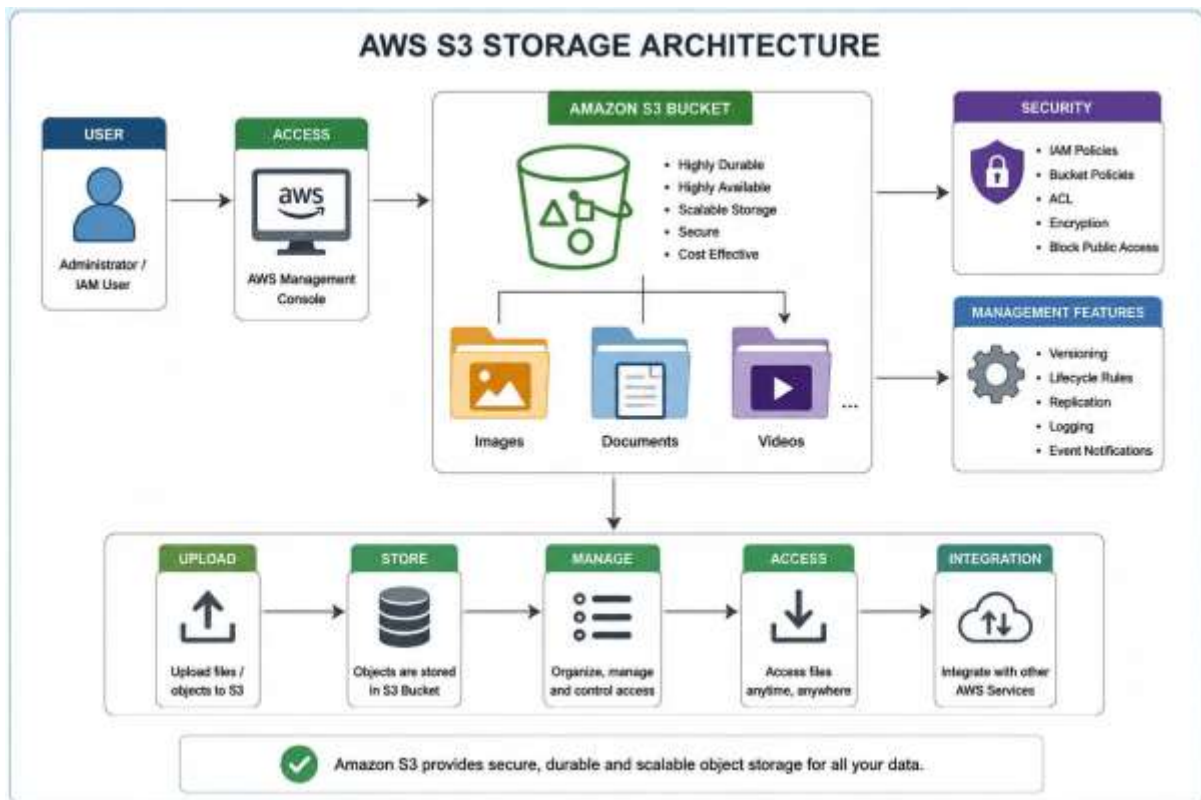


FIG-3: ARCHITECTURE DIAGRAM

3. RESOURCE CREATED-

Step 1: Create Bucket Page:

- Open Amazon S3 Dashboard.
- Click Create Bucket

- Enter:
- 1. Name:kritika-static-website
- Click Create Bucket

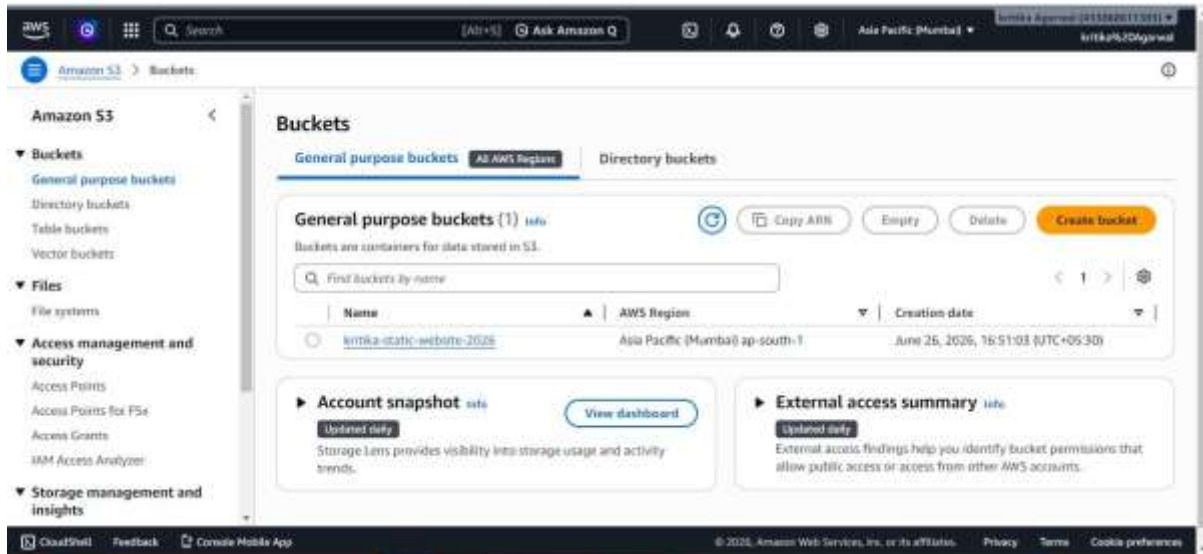


FIG: MY-BUCKET

Step 2:Create Folder Creation:

Folders (also called prefixes) are created inside the S3 bucket to organize objects in a structured way. They help separate different types of files, making data easier to manage and locate.

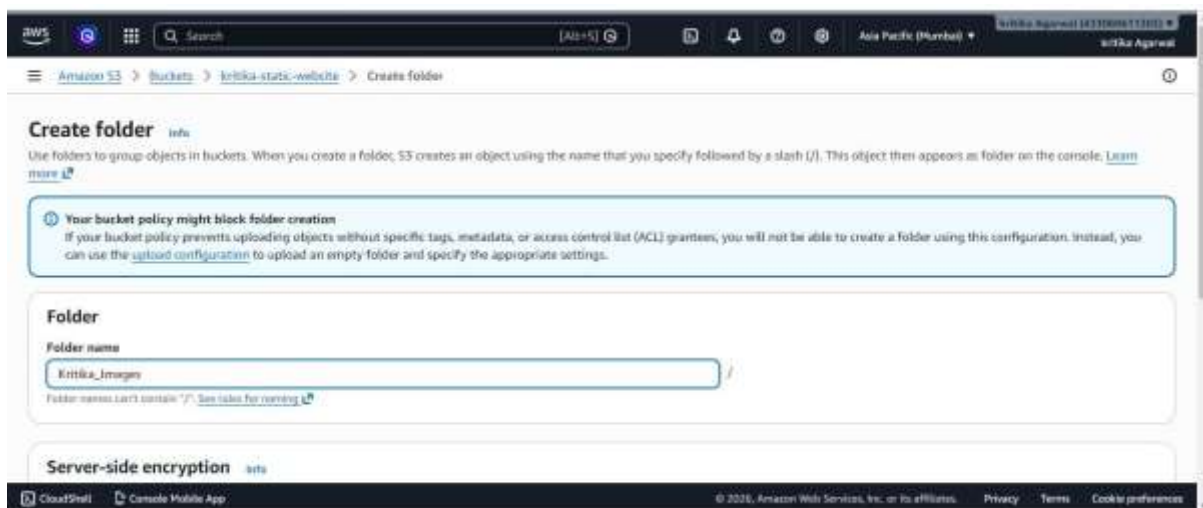


FIG:CREATE FOLDER

Step 3:File Upload:

Files are uploaded as objects into the S3 bucket. Amazon S3 supports uploading various file types such as images, PDFs, Word documents, videos, and backups. Each uploaded file becomes an object inside the bucket.

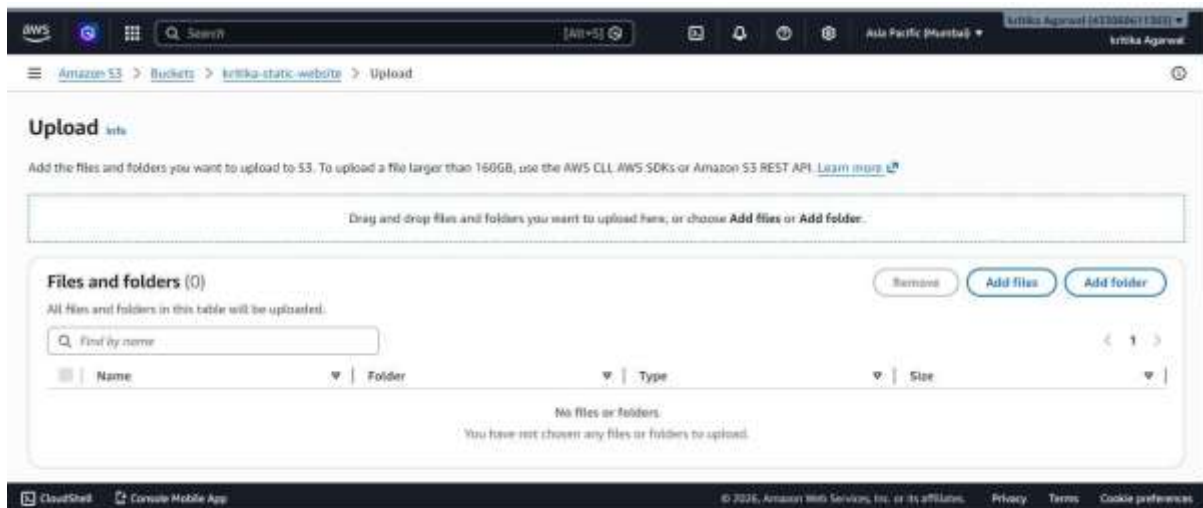


FIG: UPLOAD PAGE

Step 4: Object Management:

Object Management includes viewing, downloading, copying, moving, renaming (by copying), deleting, and managing metadata or permissions of stored objects. It helps maintain and organize files efficiently within the bucket.

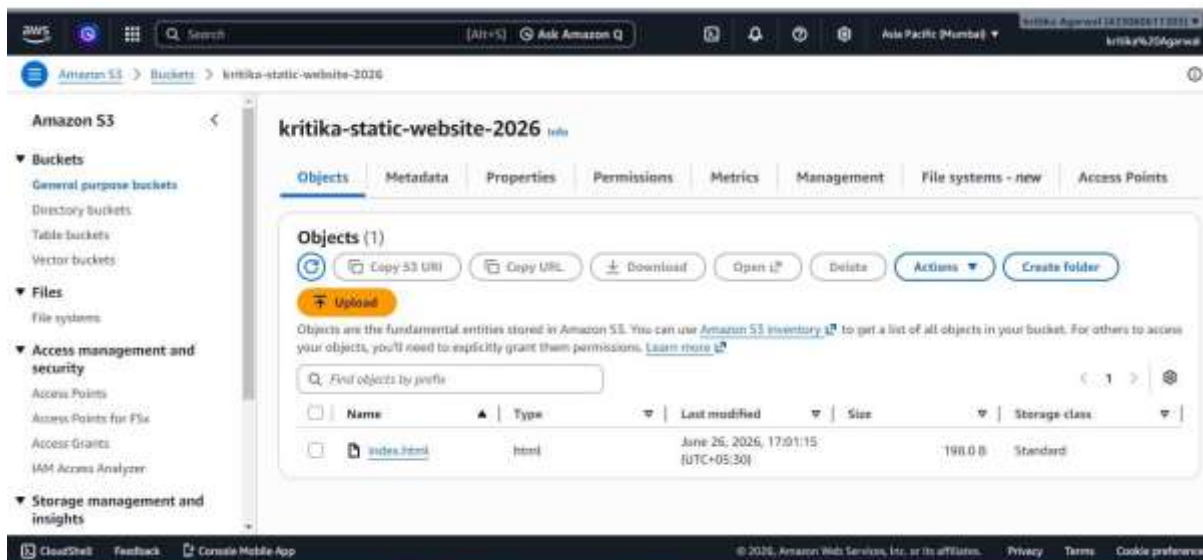


FIG: OBJECT TAB

4.CONFIGURATION DETAILS-

1. Bucket Name:

Bucket Name is the unique name assigned to an Amazon S3 bucket. Every bucket name must be globally unique because it is used to identify the bucket across all AWS accounts. The bucket name cannot be changed after the bucket is created.

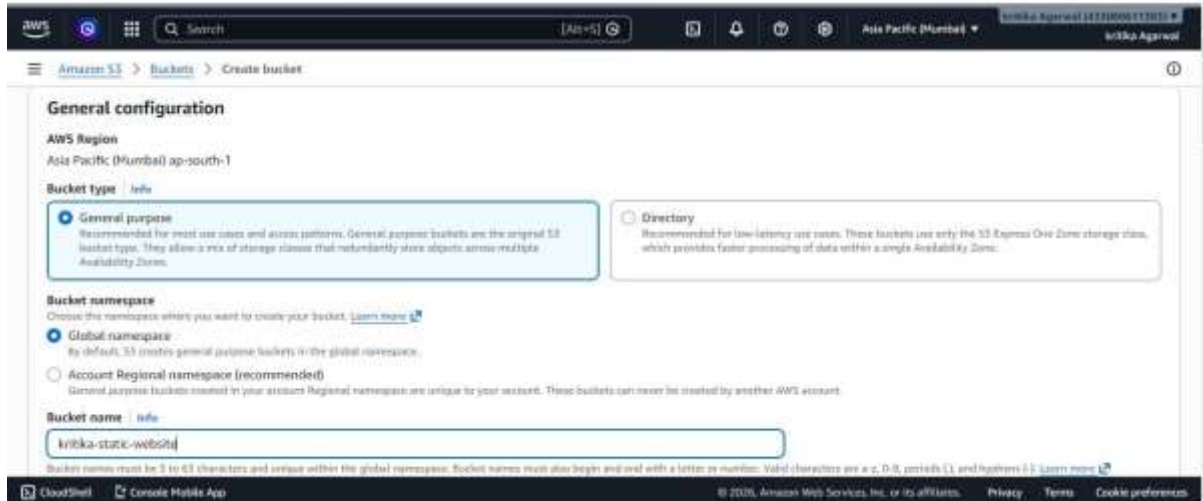


FIG: BUCKET NAME

2. AWS Region:

AWS Region specifies the geographical location where the S3 bucket is created. Choosing a region close to users reduces latency and improves performance. All objects stored in the bucket remain in the selected region unless replication is configured.

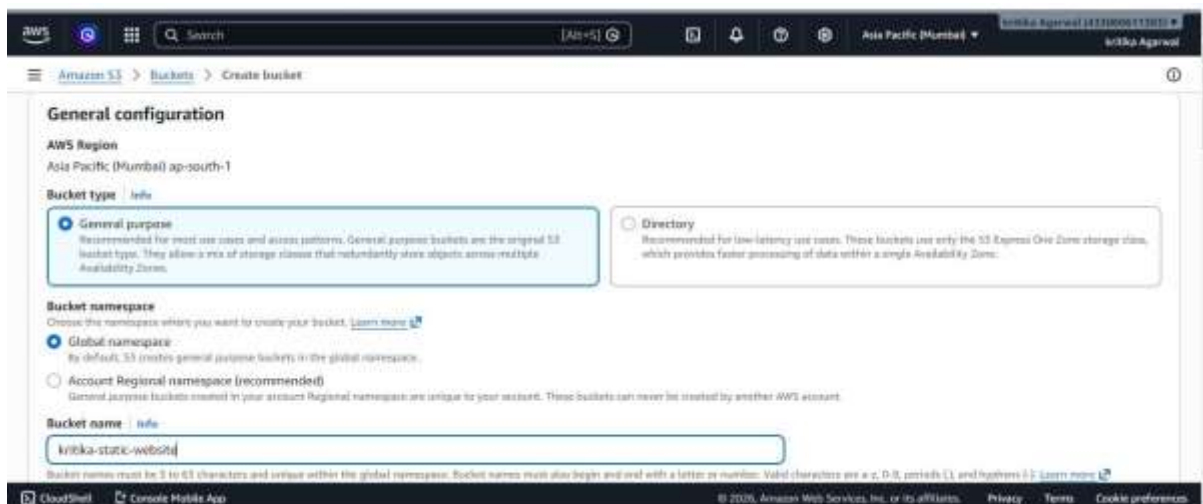


FIG: ASIA SPECIFIC REGION

3.Block Public Access:

Block Public Access is a security feature that prevents unauthorized public access to the S3 bucket and its objects. It helps protect sensitive data from being accessed over the internet.

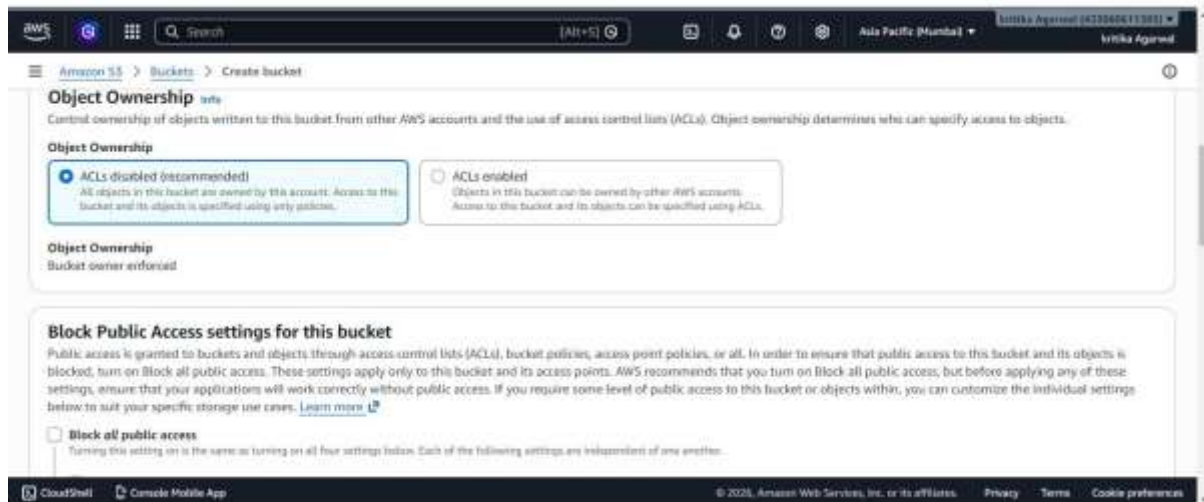


FIG: BLOCK PUBLIC ACCESS

4. Versioning:

Versioning allows Amazon S3 to keep multiple versions of the same object. If a file is accidentally deleted or overwritten, previous versions can be restored.

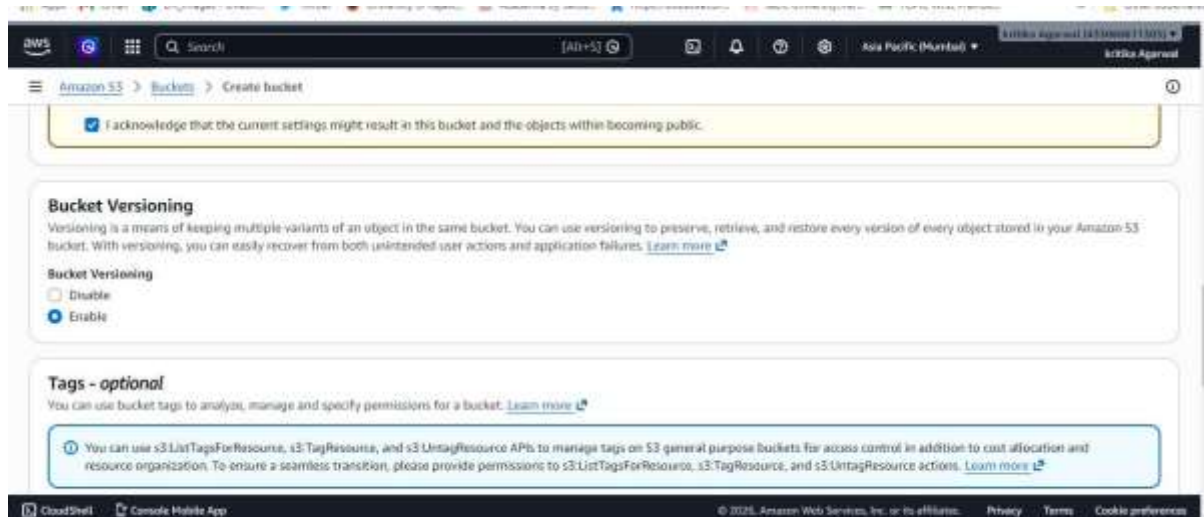


FIG: BUCKET VERSIONING

5.Default Encryption:

Default Encryption automatically encrypts all objects stored in the bucket. It protects data at rest using AWS-managed or customer-managed encryption keys.

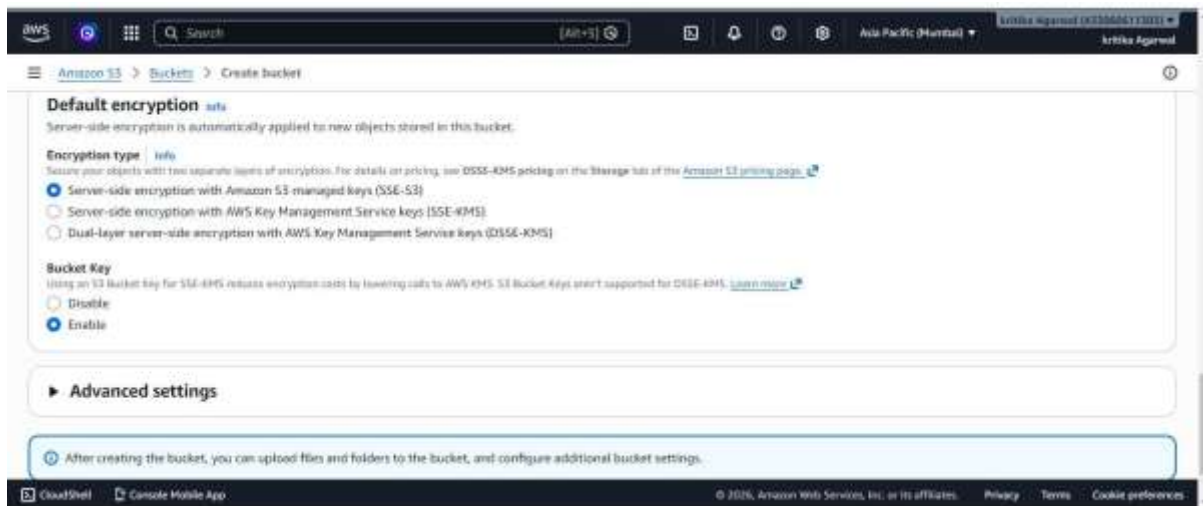


FIG: DEFAULT ENCRYPTION

6.Tags (optional):

Tags are key-value pairs used to organize, identify, and manage AWS resources. They are useful for cost allocation, resource tracking, and project management.

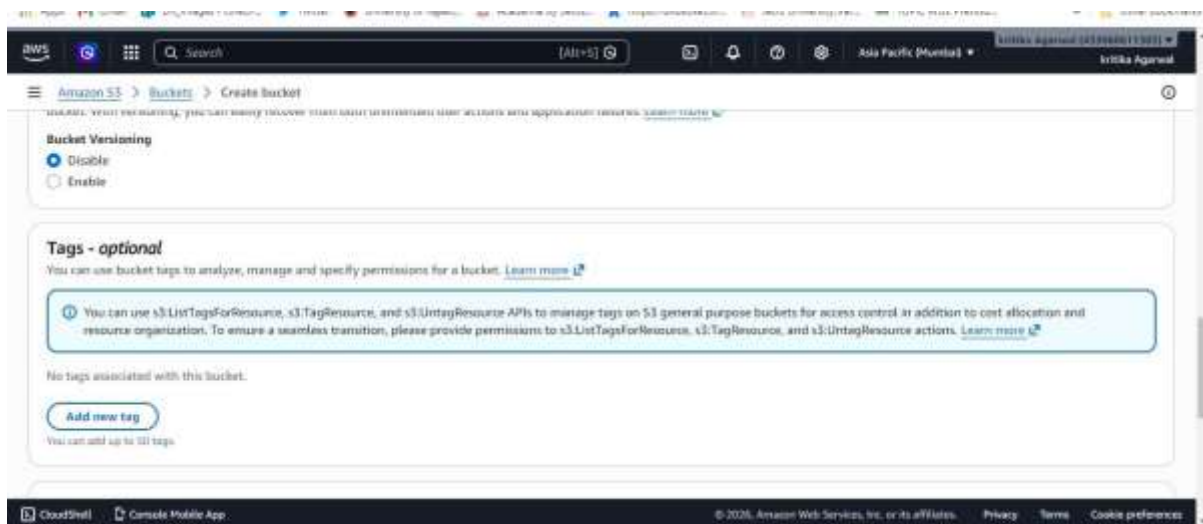


FIG: TAGS SECTION

5. SECURITY RULES-

1: IAM User / Policy:

IAM (Identity and Access Management) is used to control who can access the S3 bucket and what actions they can perform. Policies define permissions such as creating buckets, uploading files, downloading objects, and deleting data. Using IAM improves security by granting only the required permissions.

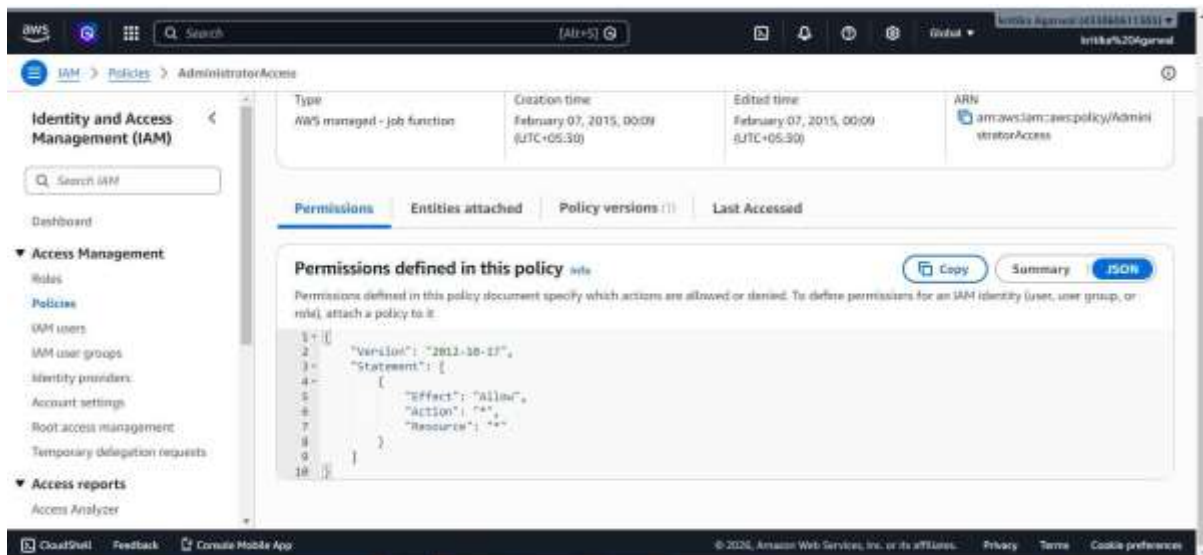


FIG:IAM USER PERMISSION

2: Bucket Policy:

A Bucket Policy is a JSON-based access policy attached directly to an S3 bucket. It controls who can access the bucket and what actions are allowed or denied. Bucket policies are commonly used to grant public or cross-account access.

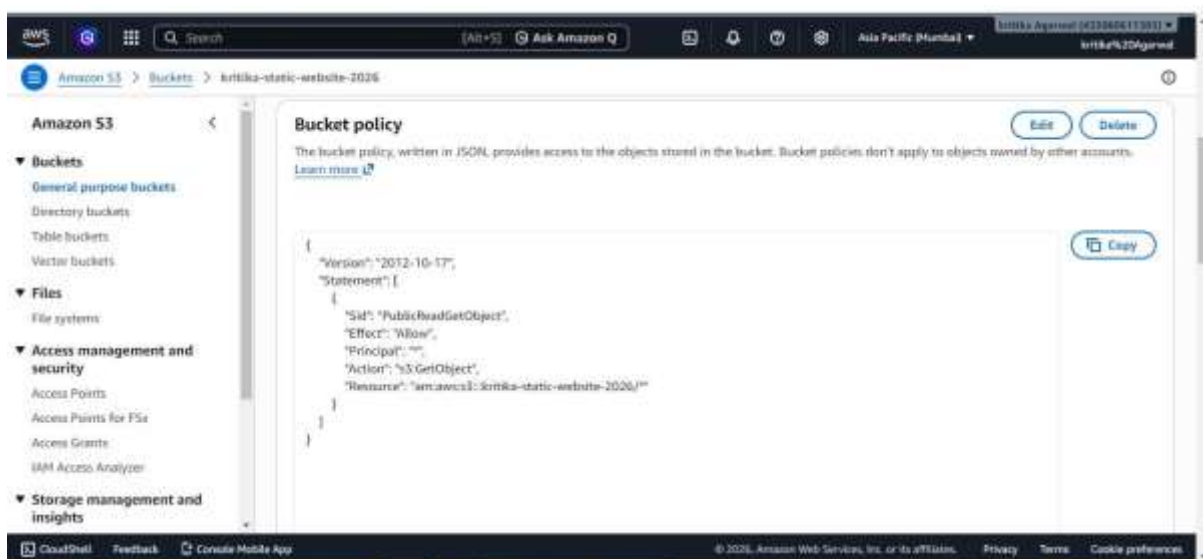


FIG:S3 BUCKET PERMISSION

3:ACL (Access Control List):

An Access Control List (ACL) provides basic read and write permissions for buckets and objects. It specifies which AWS accounts or predefined groups can access the bucket or its objects. In most modern AWS environments, IAM Policies and Bucket Policies are preferred over ACLs.

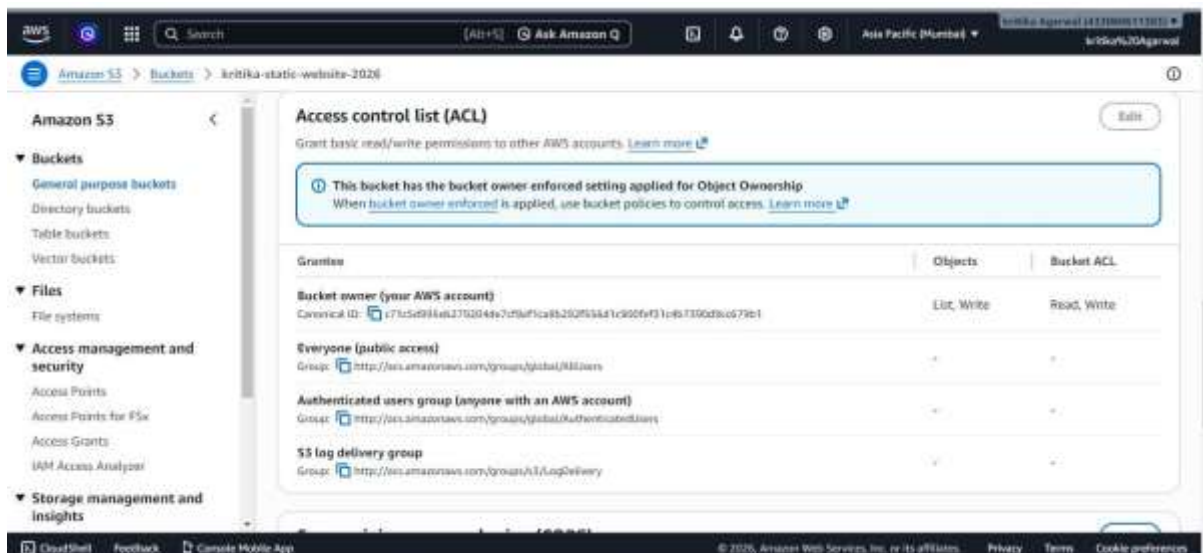


FIG: ACL PERMISSION

4:Object Ownership:

Object Ownership determines who owns the objects uploaded to the bucket. Enabling Bucket Owner Enforced ensures that the bucket owner automatically owns all uploaded objects and disables ACLs, making permission management simpler and more secure.



FIG:OBJECT OWNERSHIP

6.STORAGE MANAGEMENT-

1: Upload Objects:

Amazon S3 allows users to upload files such as images, documents, videos, and backups into a bucket. Each uploaded file is stored as an object and can be accessed according to the assigned permissions.

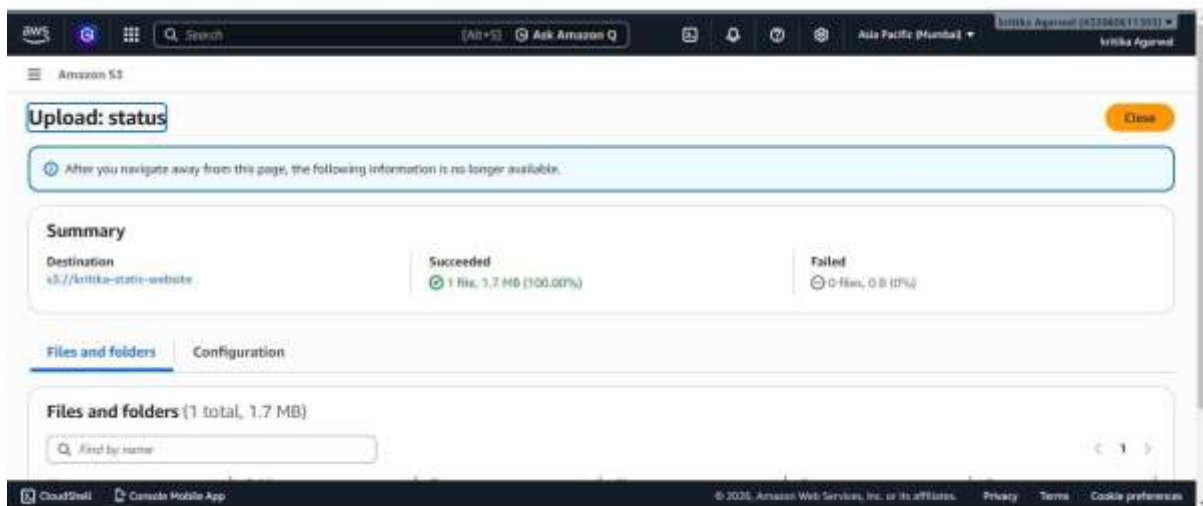


FIG:UPLOAD OBJECT

2:Download Objects:

Objects stored in an S3 bucket can be downloaded anytime by users who have the required permissions. This feature makes data retrieval quick and reliable.

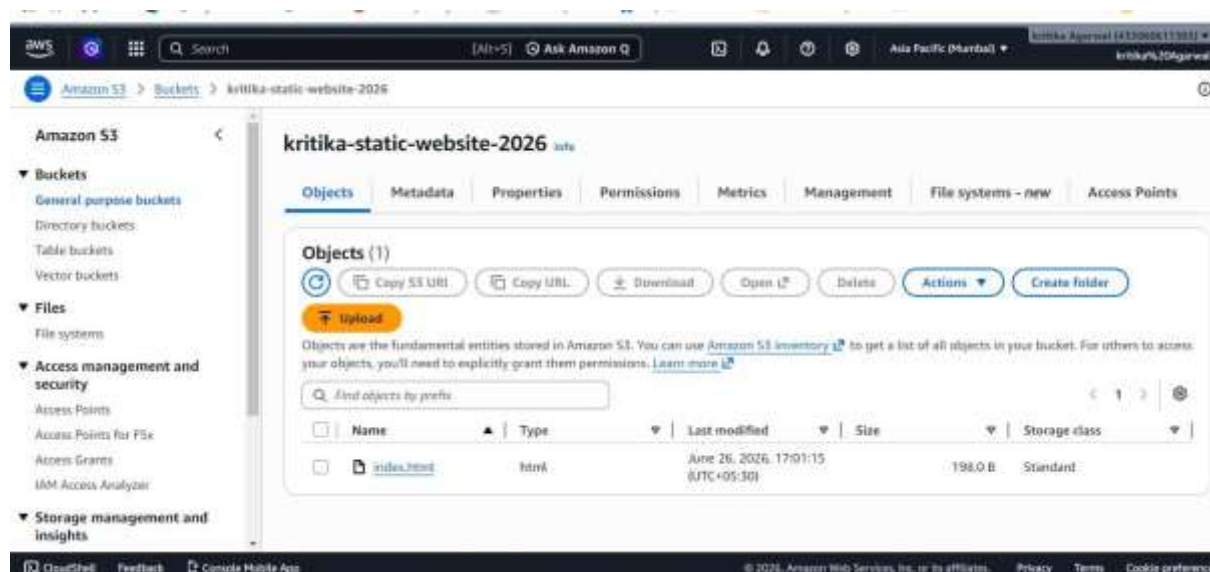


FIG:DOWNLOAD OBJECTS

3:Delete Object:

Amazon S3 allows users to permanently delete objects that are no longer needed. If Versioning is enabled, deleted objects can be recovered by restoring previous versions.

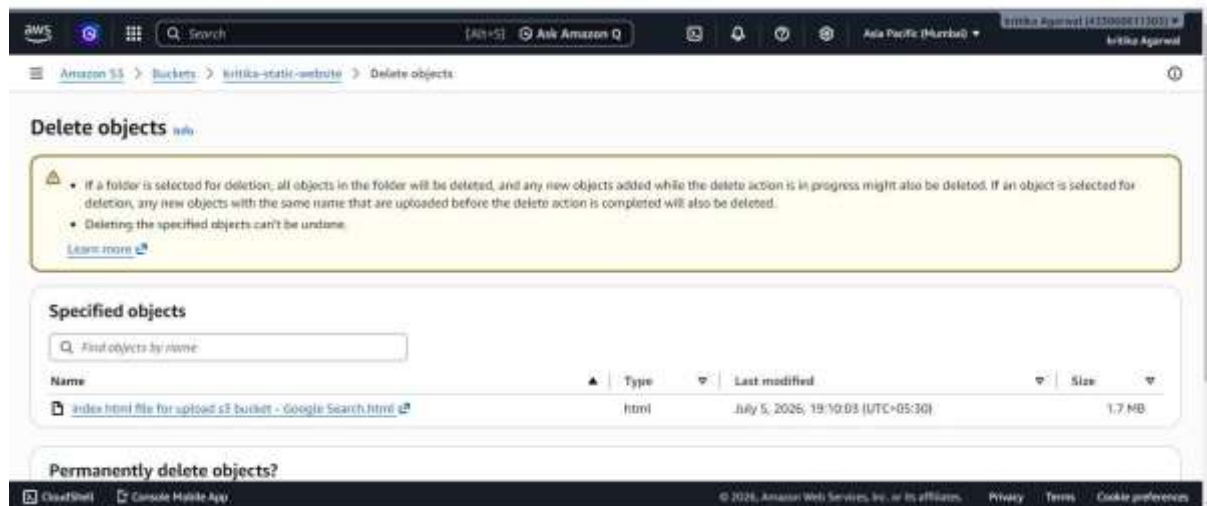


FIG:DELETE OBJECTS

4: Folder Management:

Folders (Prefixes) help organize objects inside an S3 bucket. Users can create, rename (by copying), and delete folders to keep data structured and easy to manage.

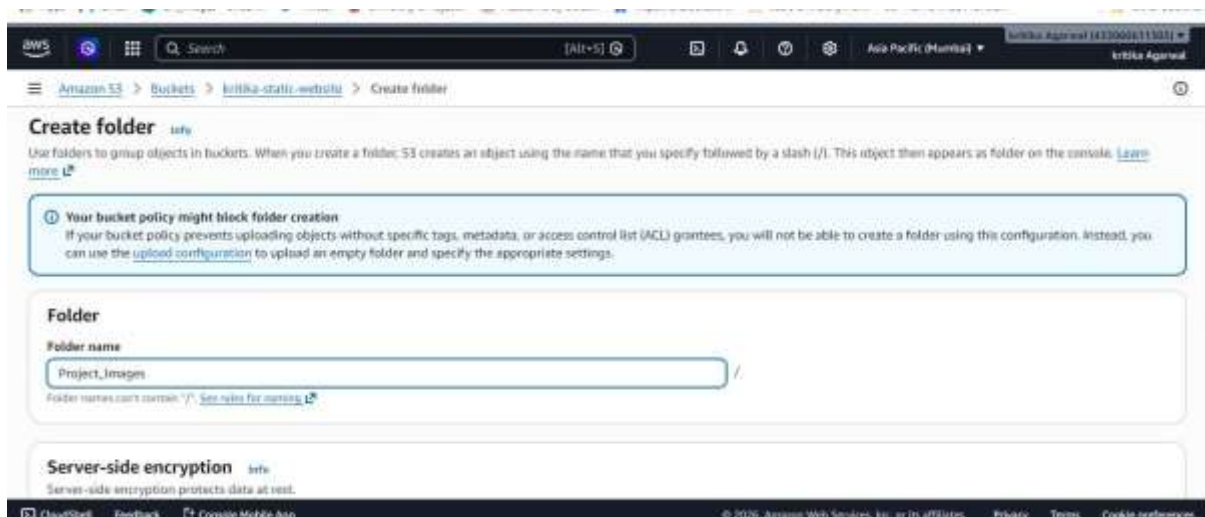


FIG:CREATE FOLDER

5: Storage Classes:

Amazon S3 provides different storage classes to optimize cost and performance based on how frequently data is accessed. Examples include S3 Standard, S3 Intelligent-Tiering, S3 Standard-IA, and Glacier.

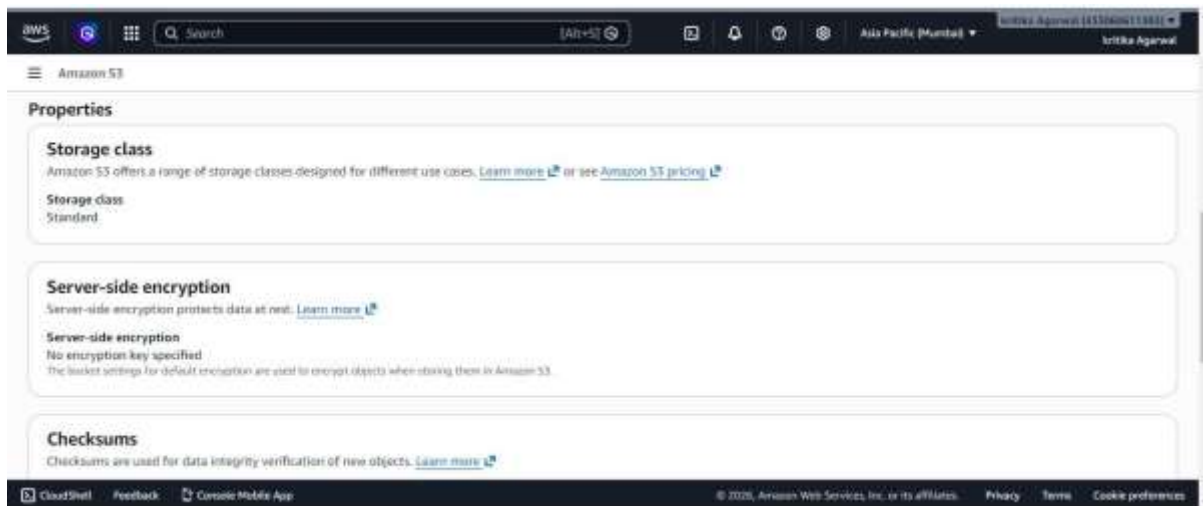


FIG:STORAGE CLASS

7.DATA PROTECTION-

1: Versioning:

Versioning stores multiple versions of an object in the same bucket. It helps recover accidentally deleted or overwritten files.

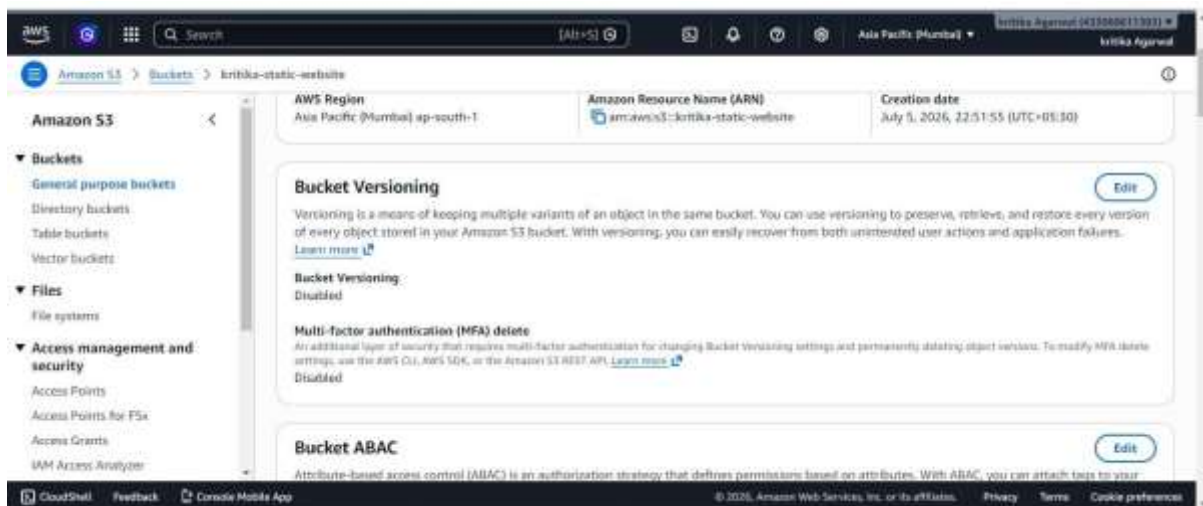


FIG:VERSIONING

2: Lifecycle Configuration:

Lifecycle Rules automatically move objects to lower-cost storage classes or delete them after a specified period, helping reduce storage costs

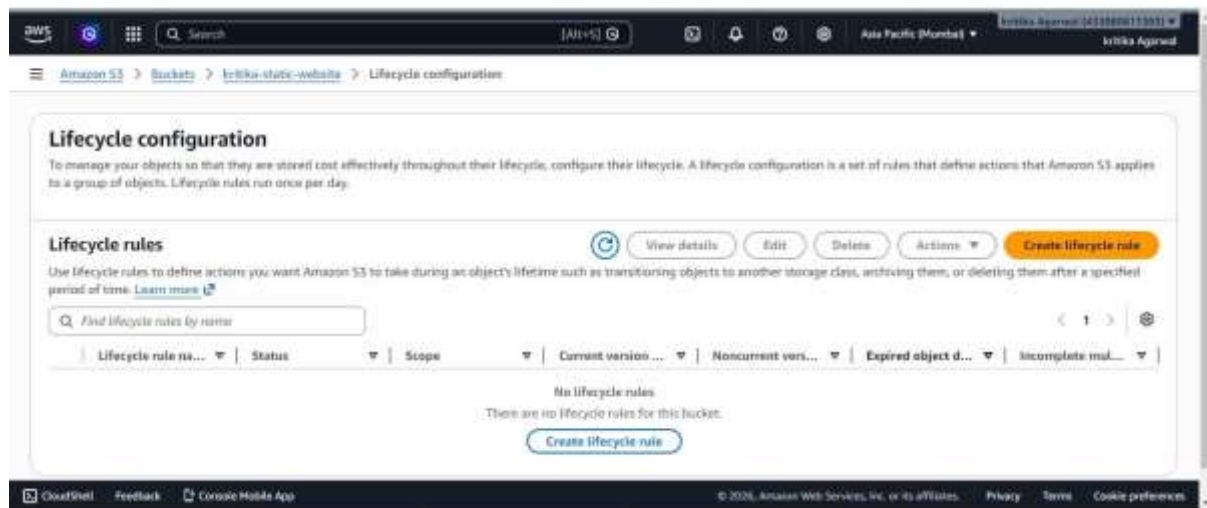


FIG: LIFECYCLE RULES

3. Replication (CRR/SRR):

Replication automatically copies objects from one bucket to another. Cross-Region Replication (CRR) copies data to another AWS Region, while Same-Region Replication (SRR) copies data within the same Region.

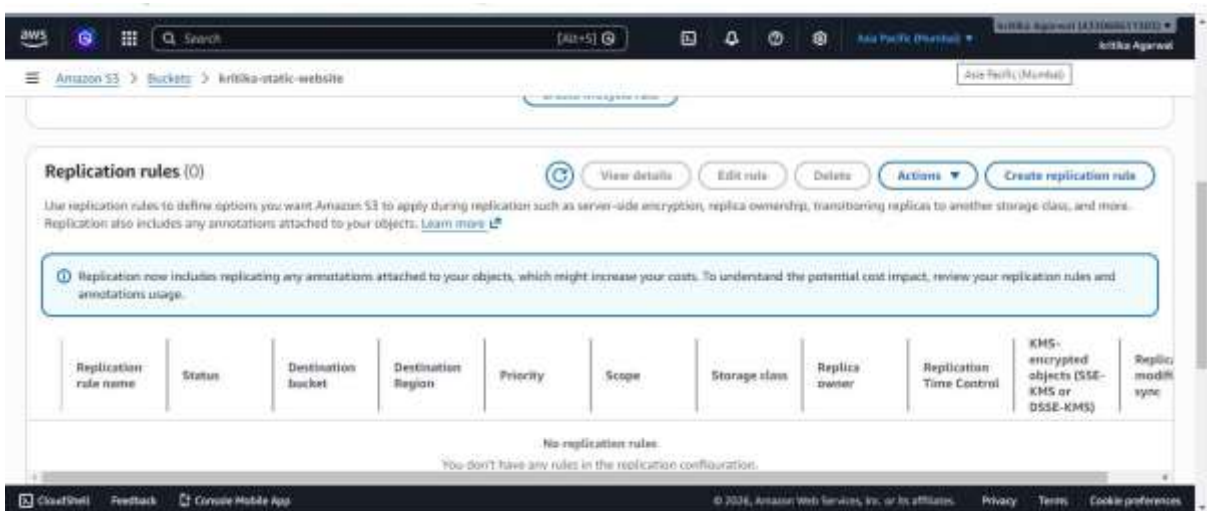


FIG: REPLICATION RULES

4. Backup or Recovery:

Amazon S3 provides reliable backup and recovery by storing data securely with high durability. Features like Versioning, Replication, and Lifecycle Rules help protect data from accidental deletion, overwriting, or data loss. If Versioning is enabled, previous versions of an object can be restored, making data recovery simple and efficient.

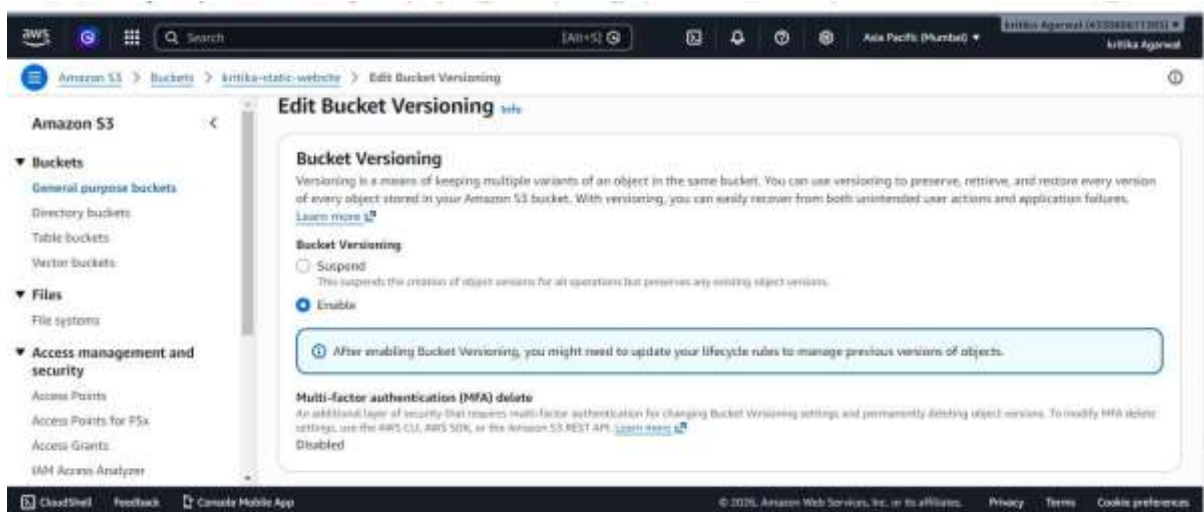


FIG-1:VERSIONING (ENABLED)

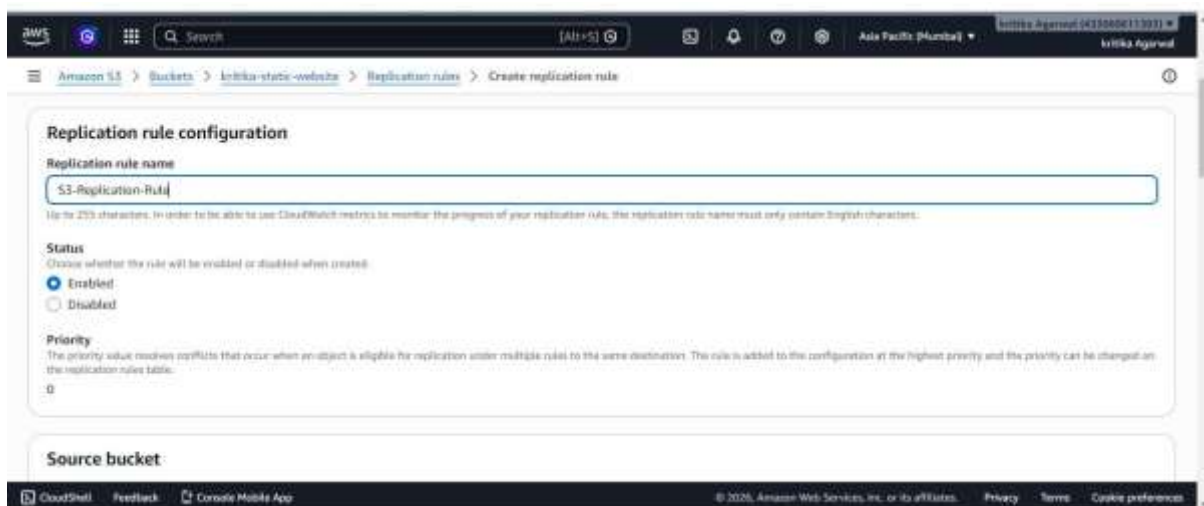


FIG-2:REPLICATION

8.TESTING RESULT-

1. Upload File Test:

A sample file was successfully uploaded to the S3 bucket, confirming that the bucket accepts object uploads correctly.

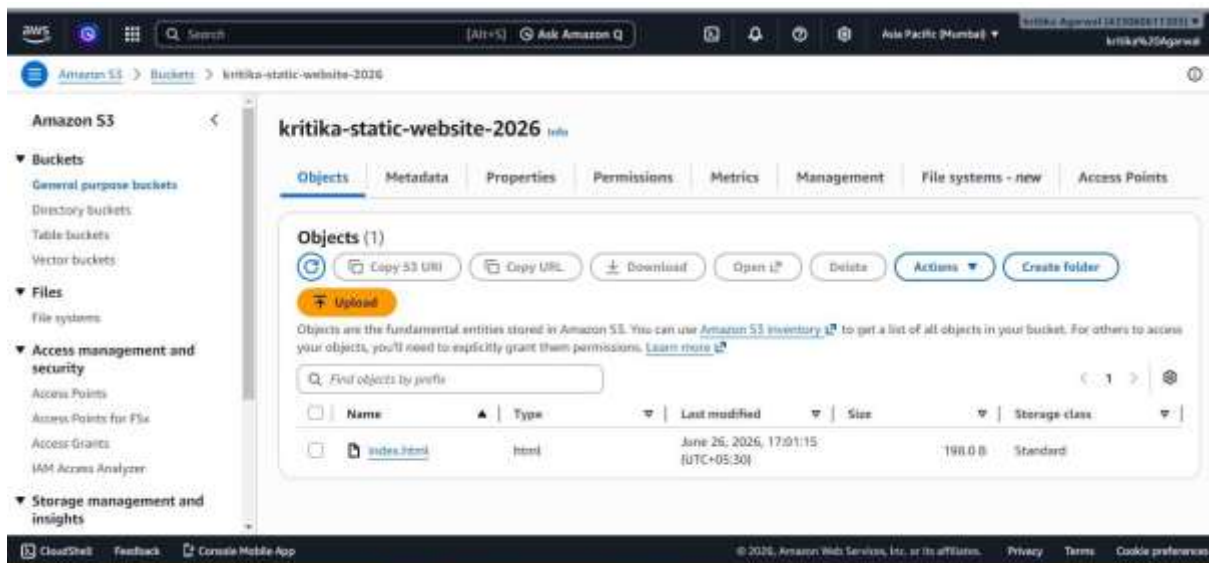


FIG: UPLOAD FILE VISIBLE

2.Download File Test:

The uploaded file was downloaded successfully from the S3 bucket, confirming that stored objects can be accessed and retrieved.

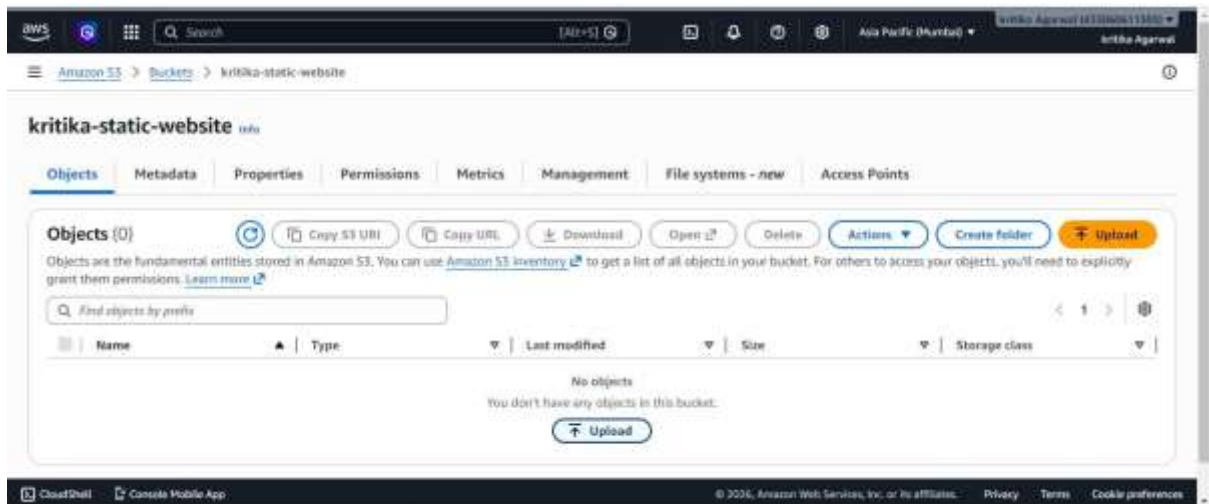


FIG: DOWNLOAD

3.Public Access Test:

The bucket's public access settings were verified to ensure that objects are accessible only according to the configured permissions.

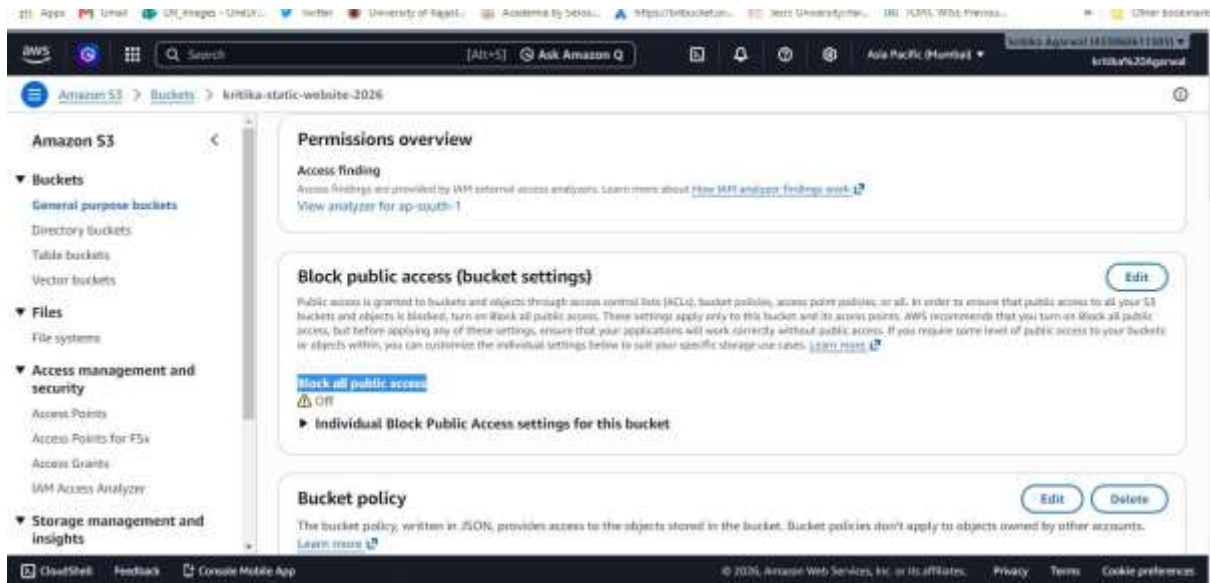


FIG: BLOCK PUBLIC ACCESS

4. Bucket Policy Test:

The bucket policy was tested to verify that only authorized users or services can access the bucket based on defined permissions.

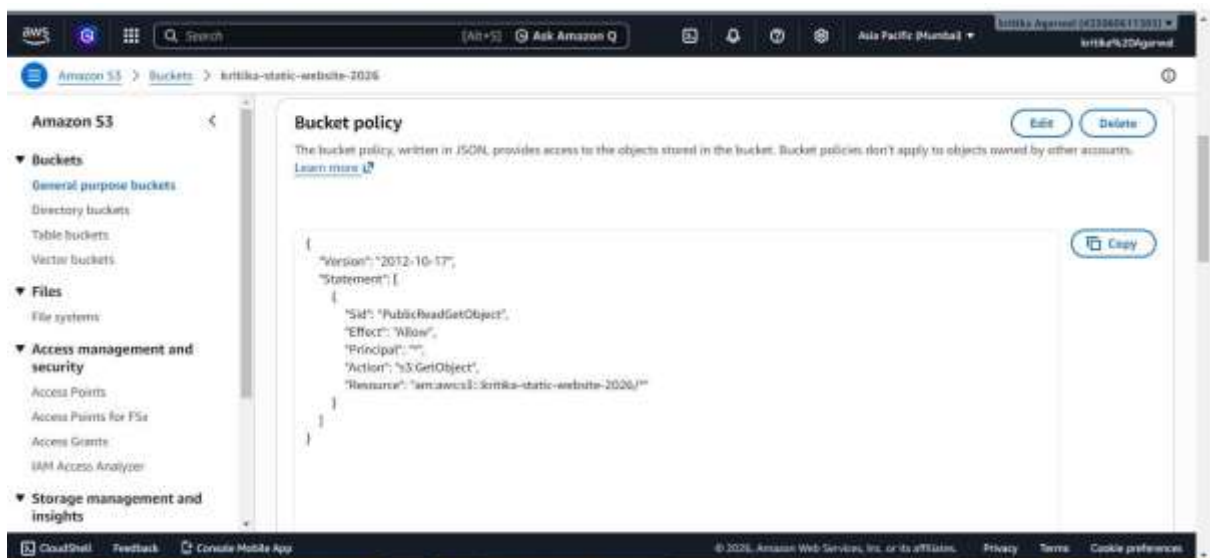


FIG: BUCKET POLICY

5. Bucket Created Successfully:

The S3 bucket was created successfully with the required configuration and was ready to store data.

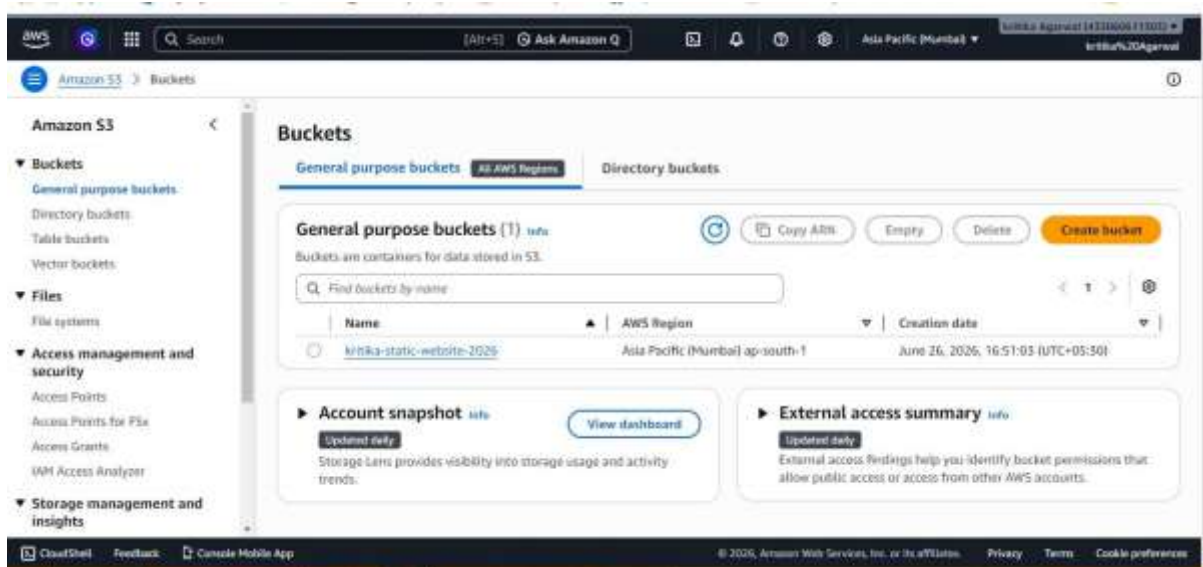


FIG:S3 BUCKET LIST

6.Files Uploaded Successfully:

Multiple files were uploaded successfully into the bucket without any errors, confirming proper storage functionality.

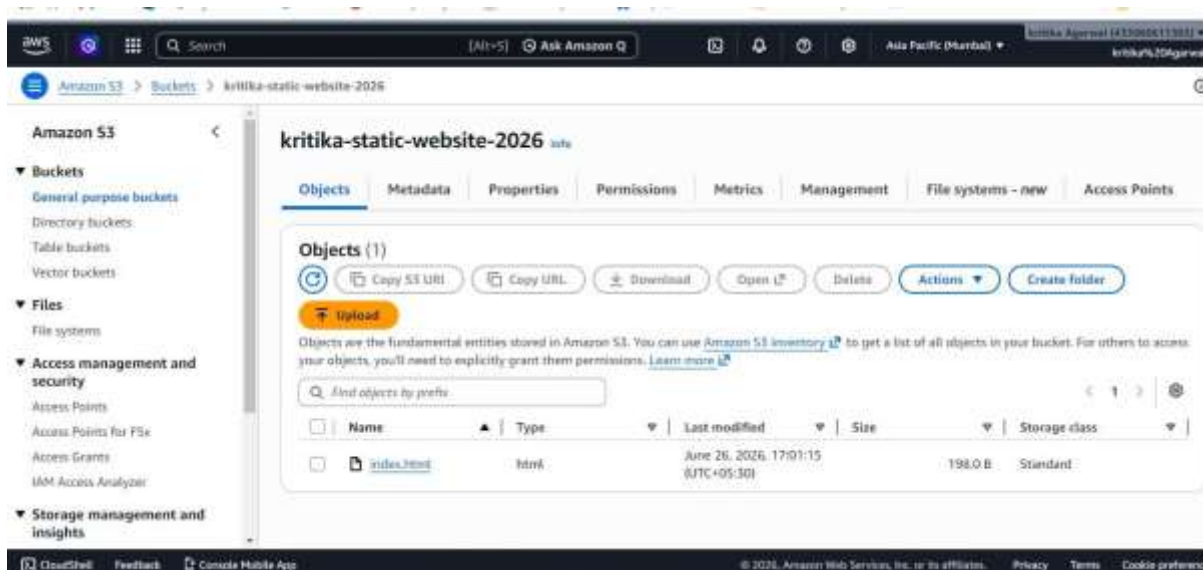


FIG:OBJECTS (UPLOADED FILES)

7.Access Verified:

Access permissions were verified by performing upload, download, and view operations using the configured IAM policy and bucket settings.

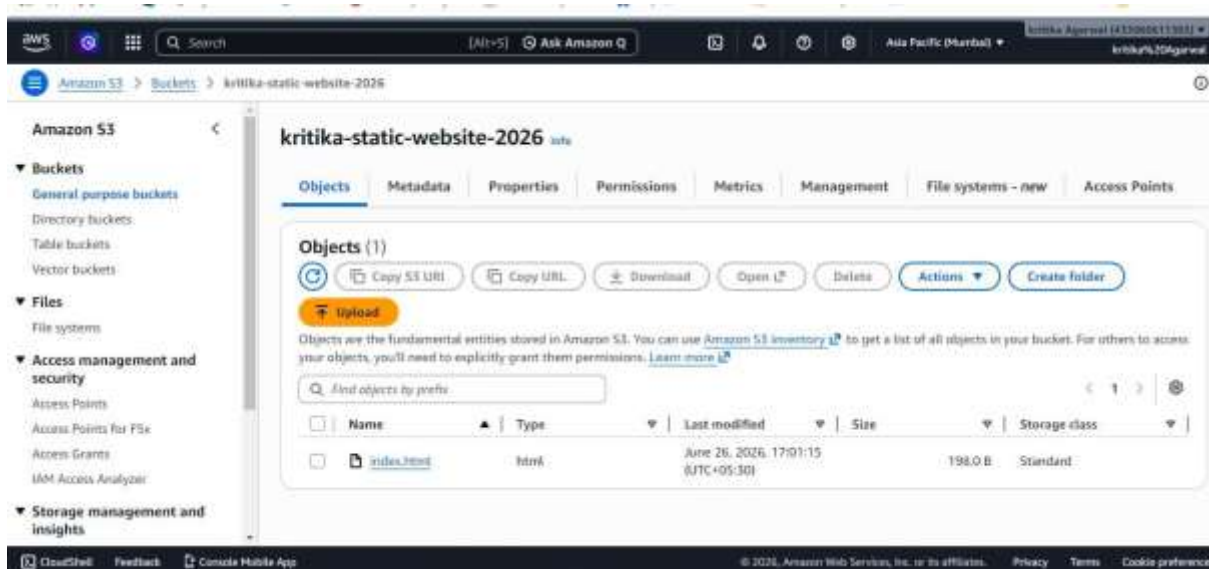


FIG:SUCCESSFUL ACCESS

9.CONCLUSION-

Project Summary:

- AWS Console login completed.
- Amazon S3 bucket created successfully.
- Folders and files uploaded.
- Security settings configured.
- Versioning and Lifecycle enabled.
- Data stored and managed successfully.
- Project tested successfully.

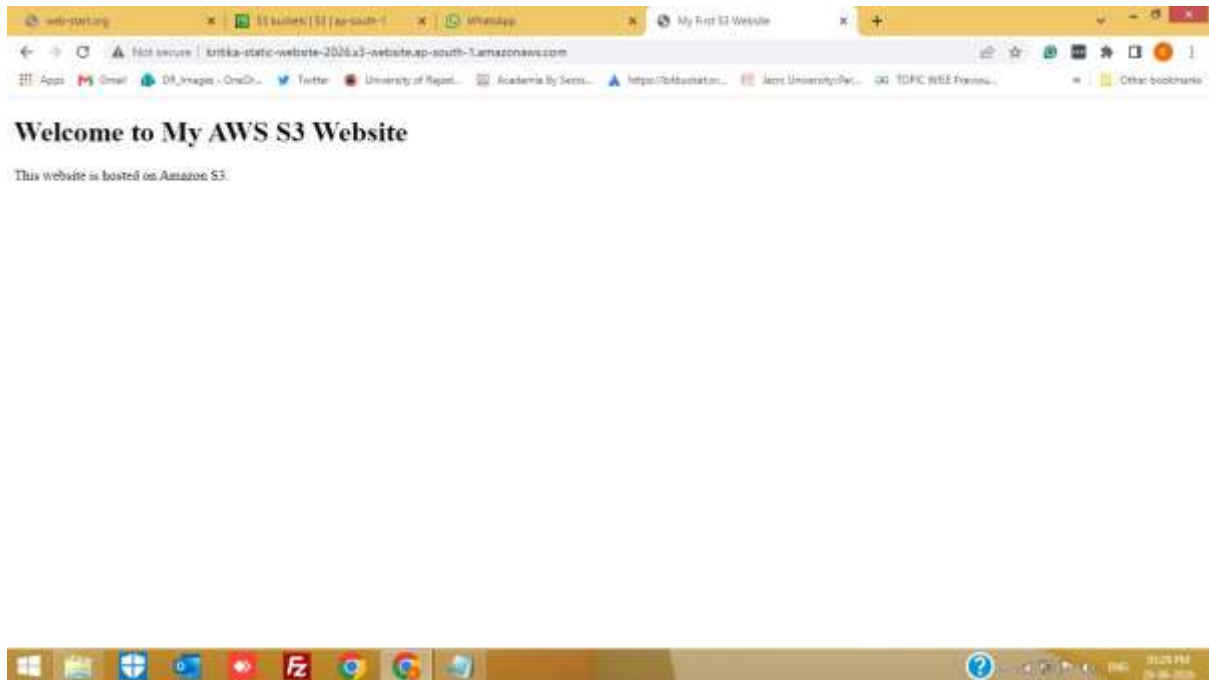


FIG:AMAZON S3

Learning Outcomes:

- Learned Amazon S3 bucket creation.
- Learned file upload and download.
- Learned folder management.
- Learned IAM and Bucket Policy.
- Gained hands-on experience with AWS S3.

THANK YOU
